

Paper 20.25 - Layer-2 Synthesis Ledger Toolkit

CQE-paper-20.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-20.25.md

Paper 20.25 - Layer-2 Synthesis Ledger Toolkit

Purpose

This supplement exposes the ledger checks for Paper 20.

Digital Route

Run:

```
python production/formal-papers/CQE-paper-20/verify_layer2_synthesis_ledger.py
```

The receipt must show ledger verification, reachability examples, transport row counts, and registry accept/reject behavior.

Analog Route

Use a four-column board:

```
solved
open
forbidden or failed
transported with residue
```

Place every row in exactly one column and write the witness on the row card.

Hidden-Guess Diagnostic

Before revealing the row receipt, classify it as:

```
solved
open
forbidden
transported
invalid promotion
```

Reveal the ledger row only after the classification.

Boundary

The toolkit may aggregate rows. It may not upgrade a row beyond the source paper's receipt.

Conclusion

Paper 20.25 makes the ledger visible as a practical audit board.